

Special Assignment

- **Name of Student** : Shruti Dnyaneshwar Darwatkar
- **Domain Name**: AIML Development
- **College Name**: Zeal Polytechnic , Narhe
- **Branch Name**: AIML
- **Github url**: https://github.com/shrutidarwatkar-sketch/Task_ITR.git

Step 1: Import Libraries

Program screenshot:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use("ggplot")
```

Step 2: Upload Dataset

Program screenshot:

```
from google.colab import files

uploaded = files.upload()
```

... Choose files Titanic-Dataset.csv
Titanic-Dataset.csv(text/csv) - 61194 bytes, last modified: 10/07/2026 - 100% done
Saving Titanic-Dataset.csv to Titanic-Dataset.csv

Step 3: Load Dataset

Program screenshot:

```
df = pd.read_csv("Titanic-Dataset.csv")
```

Step 4: Display First Few Records

Program screenshot:

```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Step 5: Check Dataset Shape

Program screenshot:

```
print("Rows and Columns:", df.shape)
```

```
... Rows and Columns: (891, 12)
```

Step 6: Display Column Names

Program screenshot:

```
print(df.columns)
```

```
... Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',  
         'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],  
         dtype='object')
```

Step 7: Check Data Types

Program screenshot:

```
df.dtypes
```

```
...      0
```

PassengerId	int64
Survived	int64
Pclass	int64
Name	object
Sex	object
Age	float64
SibSp	int64
Parch	int64
Ticket	object
Fare	float64
Cabin	object
Embarked	object

```
dtype: object
```

Step 8: Identify Missing Values

Program screenshot:

```
df.isnull().sum()
```

...	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

Step 9: Summary Statistics

Program screenshot:

```
df.describe(include='all')
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
count	891.000000	891.000000	891.000000	891	891	714.000000	891.000000	891.000000	891	891.000000	204	889
unique	NaN	NaN	NaN	891	2	NaN	NaN	NaN	681	NaN	147	3
top	NaN	NaN	NaN	Dooley, Mr. Patrick	male	NaN	NaN	NaN	347082	NaN	G6	S
freq	NaN	NaN	NaN	1	577	NaN	NaN	NaN	7	NaN	4	644
mean	446.000000	0.383838	2.308642	NaN	NaN	29.699118	0.523008	0.381594	NaN	32.204208	NaN	NaN
std	257.353842	0.486592	0.836071	NaN	NaN	14.526497	1.102743	0.806057	NaN	49.693429	NaN	NaN
min	1.000000	0.000000	1.000000	NaN	NaN	0.420000	0.000000	0.000000	NaN	0.000000	NaN	NaN
25%	223.500000	0.000000	2.000000	NaN	NaN	20.125000	0.000000	0.000000	NaN	7.910400	NaN	NaN
50%	446.000000	0.000000	3.000000	NaN	NaN	28.000000	0.000000	0.000000	NaN	14.454200	NaN	NaN
75%	668.500000	1.000000	3.000000	NaN	NaN	38.000000	1.000000	0.000000	NaN	31.000000	NaN	NaN
max	891.000000	1.000000	3.000000	NaN	NaN	80.000000	8.000000	6.000000	NaN	512.329200	NaN	NaN

Step 10: Dataset Information

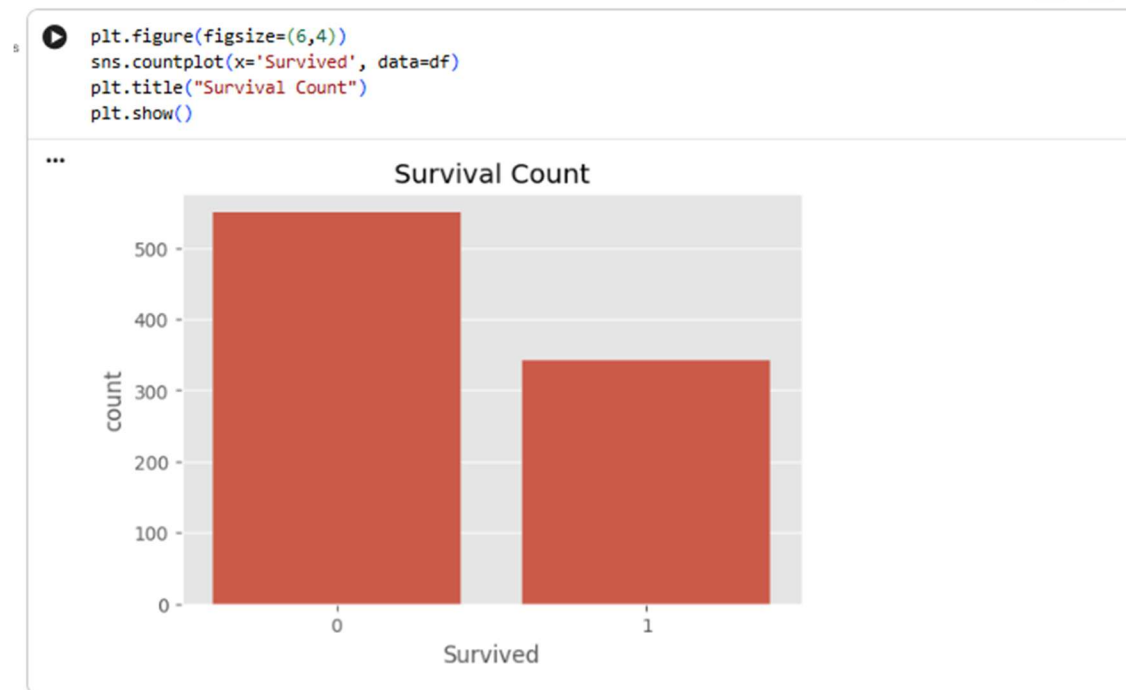
Program screenshot:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  -
 0   PassengerId    891 non-null    int64
 1   Survived       891 non-null    int64
 2   Pclass         891 non-null    int64
 3   Name           891 non-null    object
 4   Sex            891 non-null    object
 5   Age            714 non-null    float64
 6   SibSp          891 non-null    int64
 7   Parch          891 non-null    int64
 8   Ticket         891 non-null    object
 9   Fare           891 non-null    float64
10   Cabin          204 non-null    object
11   Embarked       889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

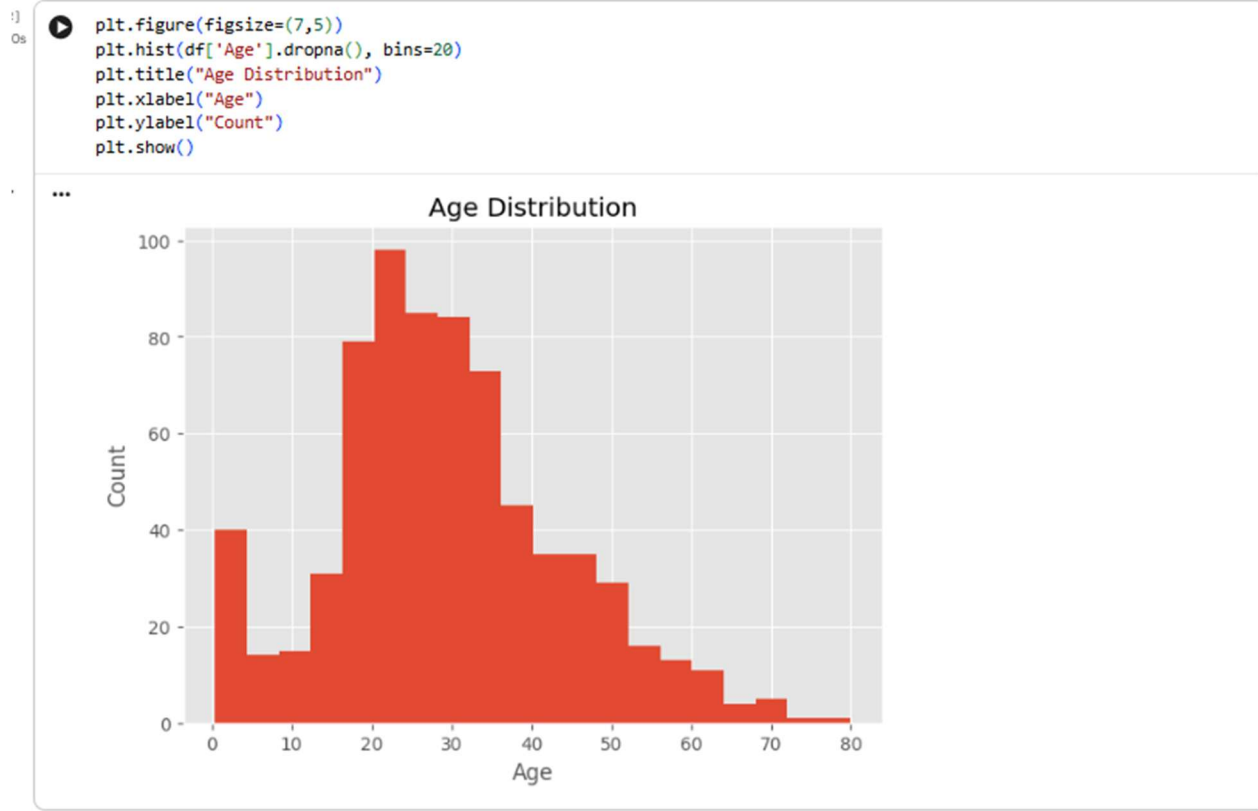
Step 11: Survival Count (Bar Chart)

Program screenshot:



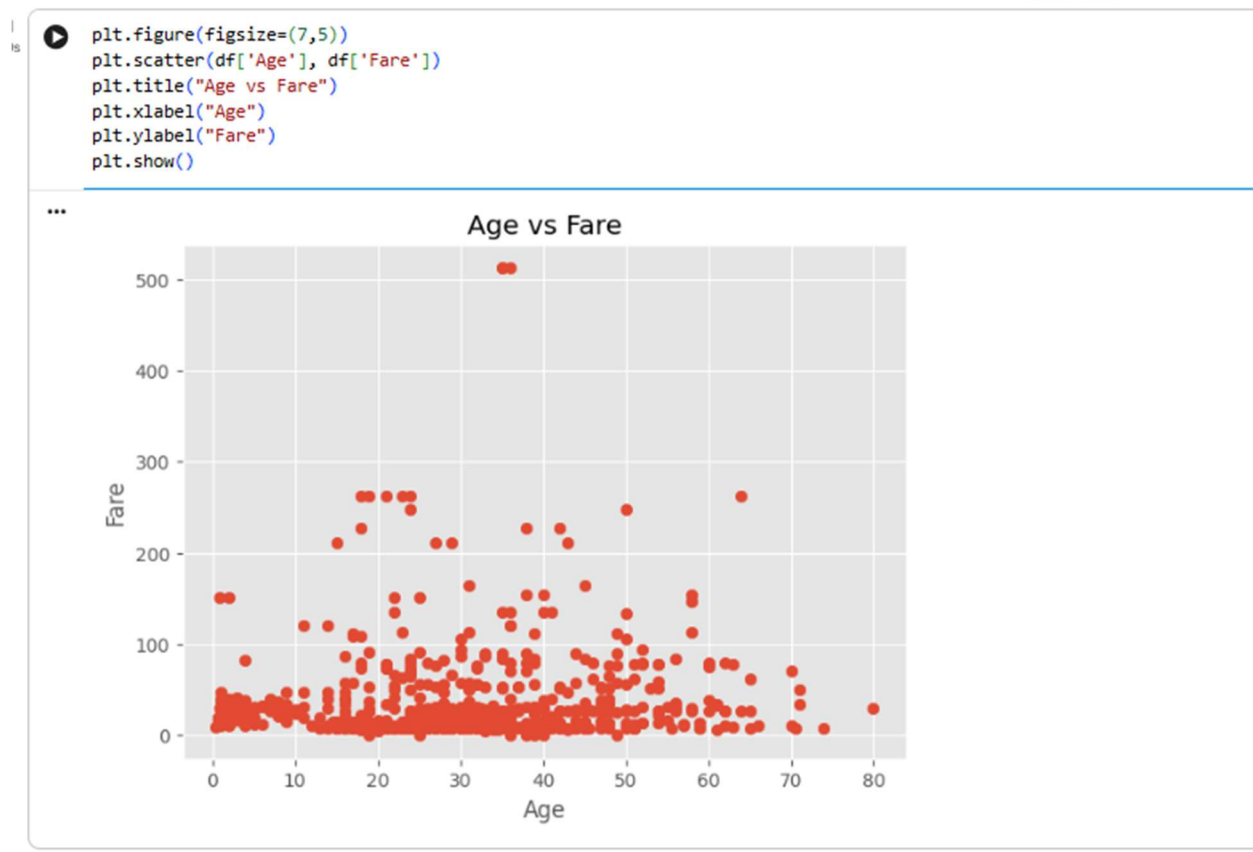
Step 12: Age Distribution (Histogram)

Program screenshot:



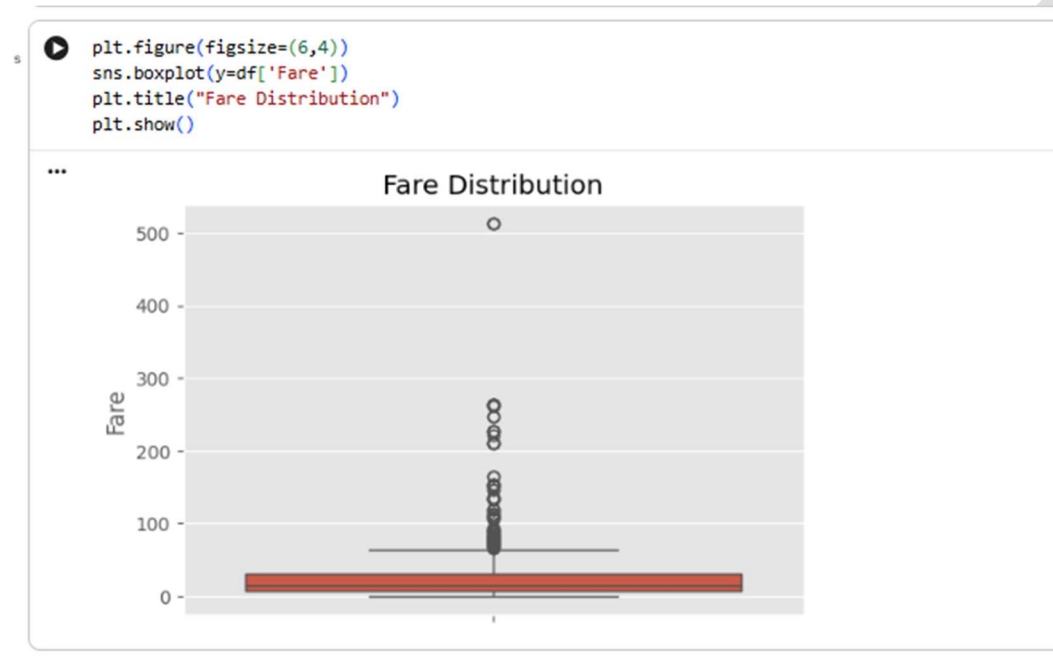
Step 13: Age vs Fare (Scatter Plot)

Program screenshot:



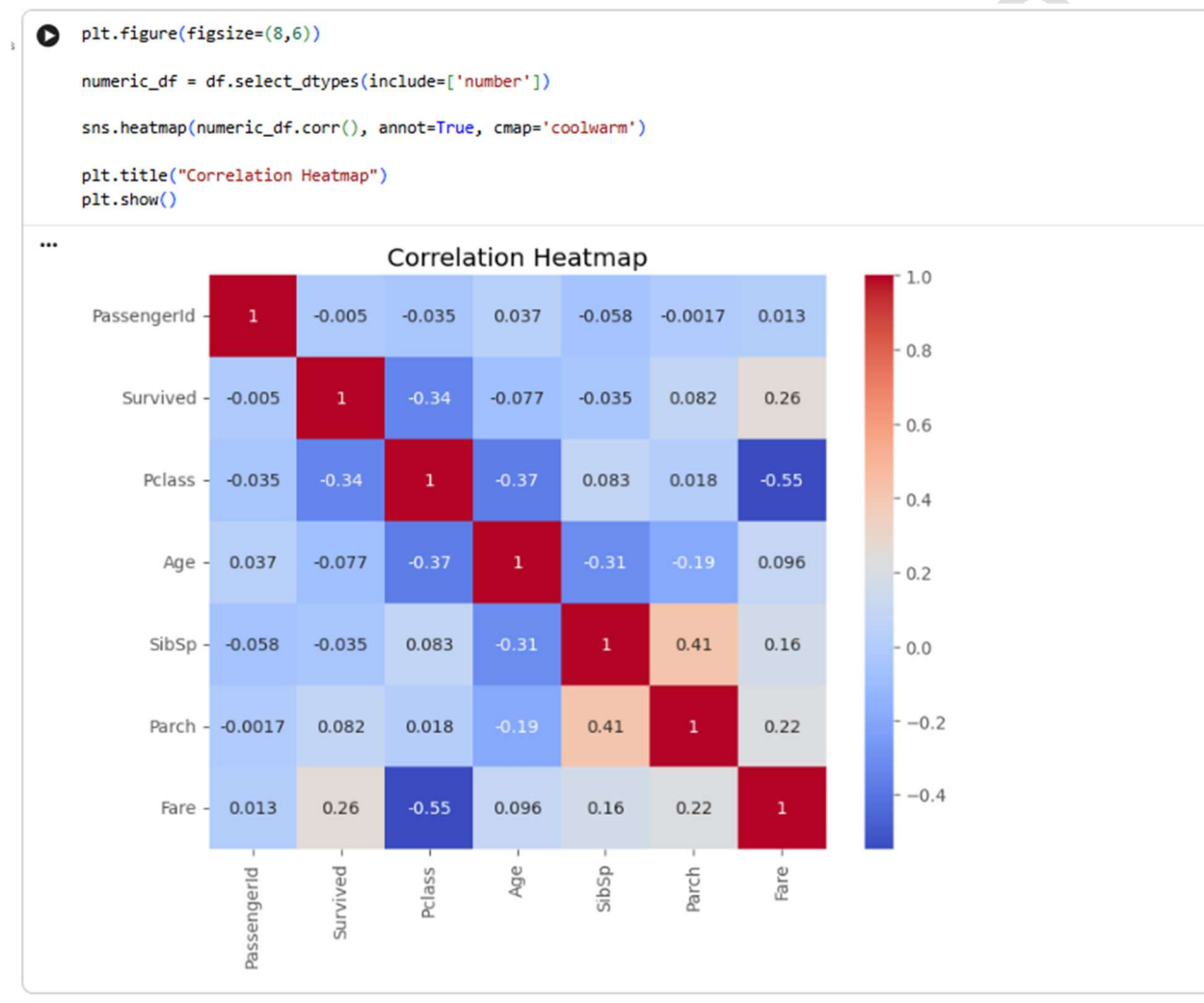
Step 14: Fare Distribution (Box Plot)

Program screenshot:



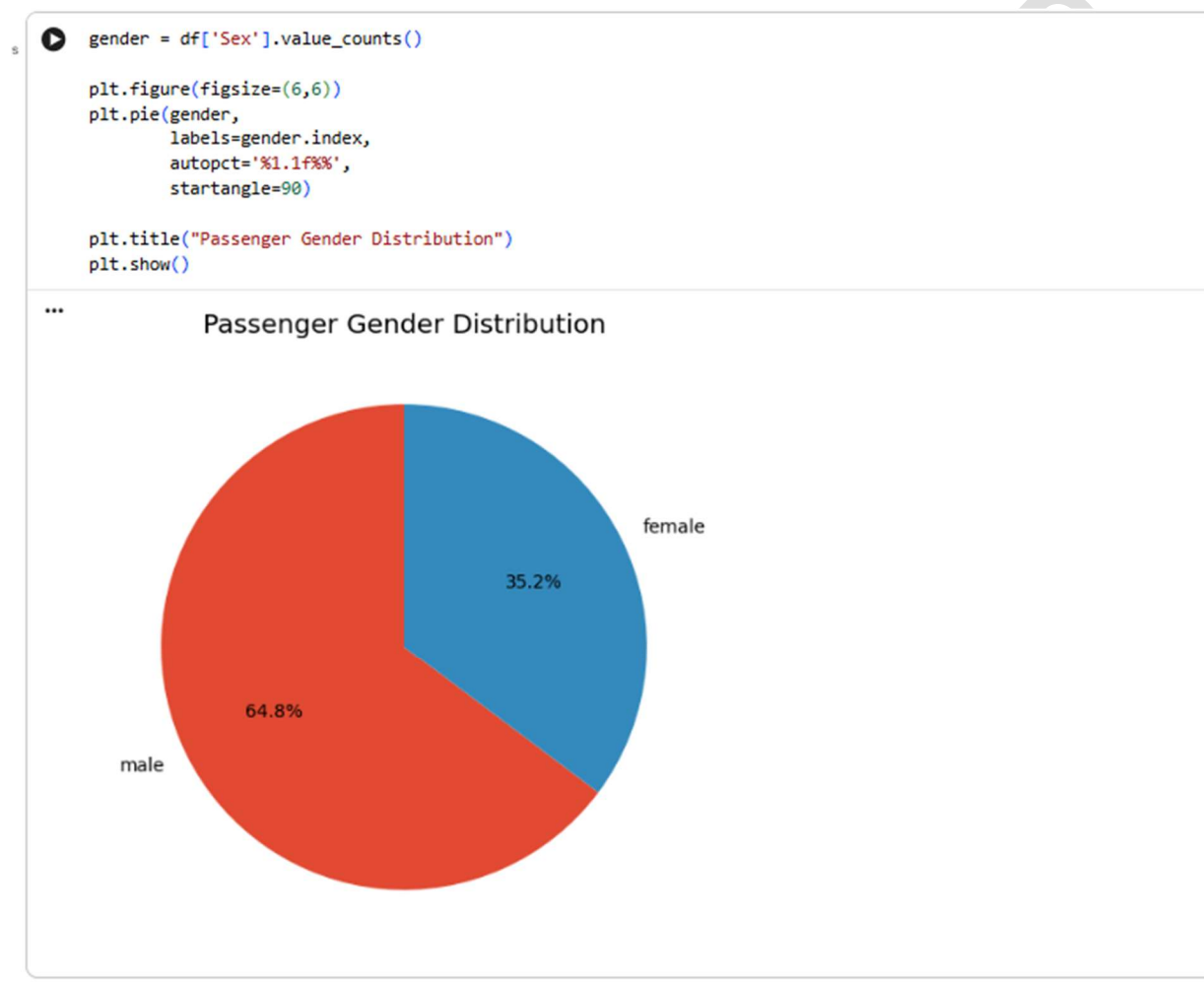
Step 15: Correlation Heatmap

Program screenshot:



Step 16: Passenger Gender (Pie Chart)

Program screenshot:



Summary:

```
"""Summary

1. The Titanic dataset contains 891 rows
   and 12 columns.

2. It includes passenger details such as
   Age, Sex, Passenger Class (Pclass),
   Fare, Cabin, Embarked, and Survived.

3. The target variable is Survived,
   making it a classification dataset.

4. Missing values are present in the
   Age, Cabin, and Embarked columns.

5. Most passengers were male.

6. The majority of passengers traveled
   in Third Class (Pclass = 3).

7. More passengers did not survive
   than survived.

8. Female passengers had a higher
   survival rate than male passengers.

9. First Class passengers had a better
   chance of survival than Second and
   Third Class passengers.

10. Most passengers were between
    20 and 40 years of age.

11. The Fare column contains a few
    outliers with very high ticket prices.

12. Passenger Class and Fare show
    a relationship with survival.

13. Missing values and outliers should
    be handled before building a
    machine learning model.

14. The dataset is suitable for
    Exploratory Data Analysis (EDA)
    and classification tasks.

Conclusion

15. Gender, passenger class, age,
    and fare are the most important
    factors affecting passenger survival.

16. After data cleaning and
    preprocessing, the dataset can be
    used to build accurate classification
    models for predicting passenger
    survival."""
```

1. Dataset Name:

Titanic Dataset

2. Kaggle Dataset Link :

<https://www.kaggle.com/datasets/yasserh/titanic-dataset>

3. Dataset Description:

The Titanic dataset contains information about passengers aboard the RMS Titanic. It includes details such as passenger class, name, gender, age, fare, ticket, cabin, embarkation point, and whether the passenger survived. This dataset is widely used for Exploratory Data Analysis (EDA) and Machine Learning classification problems.

4. Number of Rows and Columns:

- Rows: 891
- Columns: 12

5. List of Dataset Features (Columns):

- PassengerId
- Survived
- Pclass
- Name
- Sex
- Age
- SibSp
- Parch
- Ticket
- Fare
- Cabin
- Embarked

6. Input Features (Independent Variables):

- PassengerId
- Pclass
- Name
- Sex
- Age
- SibSp
- Parch
- Ticket
- Fare
- Cabin
- Embarked

7. Output/Target Variable (Dependent Variable) :

- 0 = Did Not Survive
- 1 = Survived

8. Type of Machine Learning Problem:

Classification

9. Conclusion:

The Exploratory Data Analysis (EDA) of the Titanic dataset shows that passenger survival is influenced by factors such as gender, passenger class, age, and fare. Female passengers and first-class passengers had a higher survival rate than others. The dataset contains missing values in the Age, Cabin, and Embarked columns, which should be handled before building a machine learning model. Overall, the Titanic dataset is an excellent dataset for learning data preprocessing, visualization, feature analysis, and classification using machine learning.